DCM v6.0.0

Phase A / B / C Master Engineering Blueprint

Frozen contract for block empirical likelihood, hierarchical event-world simulation, primitive-stat ledgers, conservation identities, and exact PrizePicks 2026 settlement. Canonical v5.4.1 remains untouched. This document consolidates the design doctrine, mathematical contracts, schema freeze, basketball primitive implementation, PrizePicks rule tables, Leaderboard mathematics, and research-only challengers (Bartlett, DPMM, GP-LVM).

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Schema freeze	PHASE_BC_SCHEMA_V1_2026-08-25
Schema contract SHA-256	6e78dacc19843338643bdcabc7477fd3ce2dd065da1e9629646dacc21cdb1f22
PrizePicks rule snapshot	PRIZEPICKS_PLAYER_PICKS_2026-08-25_V1
Baseline	Canonical DCM v5.4.1 (hashes verified against install manifest)
Status of Bartlett / DPMM / GP-LVM	RESEARCH_ONLY / SHADOW_CHALLENGER — no production authority
Canonical package	UNTOUCHED
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Thesis. Simulate a sporting event once as a shared primitive world. Derive every market from that world under a versioned definition. Translate the world through a frozen platform contract. Fail closed on unknown state. Never guess an unresolved settlement. Never sample a composite independently of its primitives.

Contents

0. Global doctrine and inherited v5.4.1 constraints

v6 does not reopen Phase A production gates and does not replace the v5.4.1 event-first, unit-safe, versioned-definition architecture. It formalizes what v5.4.1 already required and adds an exact platform-economics layer that the earlier “6-leg parlay table” objective could not represent.

Four universal persistence rules

- **Frozen objects are immutable.** After a content hash is written, fields cannot be mutated.
- **Derived values never overwrite primitives.** A simulator method may not write PRA, Fantasy, or any composite into the primitive ledger.
- **Unknown state is explicit.** Never coerce unknown to false, zero, WIN, LOSS, or eligible.
- **Anything that can change a forecast or a settlement has its own version and hash.** Platform formulas, team maps, lineup assumptions, reboot tables, and payout displays included.

Every persistent object carries

```
schema_version: str
created_at_utc: datetime
learning_revision: str
source_hashes: tuple[str, ...]
content_hash: str
```

Inherited v5.4.1 invariants that remain production law

- Shared event worlds: related markets are derived from one simulated event, not independent draws.
- Conservation and cross-market identities must hold in every world, not merely in expectation.
- Exact platform definitions: (Platform, League, Market, DefinitionVersion) with no fuzzy fallback.
- Unresolved settlements fail closed. They are not guessed.
- GOBLIN / discounted selections may be analyzed and settled; they remain forbidden as DCM selection targets.
- Primitive-stat validity does not automatically make a PrizePicks market selectable.

Separation of concerns is the load-bearing wall of v6: sports prediction produces a shared primitive world; the settlement engine translates that world through the exact frozen platform contract. Those two systems must not contaminate each other.

1. Phase A — BEL, Bartlett correction, fixed-b, BlockPlan

Bartlett correction is a higher-order research refinement, not a production gate. Block-length selection is a frozen policy object, not a claimed universal optimum. Fixed-b remains the main temporal-robustness challenger. Convex-hull failure is a separate problem that no Bartlett factor solves.

1.1 Estimating-equation contract (chronological slate)

For chronologically ordered slate-level estimating equations, construct overlapping (or non-overlapping) blocks of length L beginning every S observations. The block estimating function is the block sum of the observation-level scores. For DCM’s score-difference mean the relevant score is the centered contrast of the target parameter. The block empirical likelihood (BEL) is the usual profile EL on the blocked scores, subject to the moment constraint that the weighted block scores sum to zero, with Lagrange multiplier λ solving the first-order condition under the unit-simplex weights.

Under the ordinary small-b regime the uncorrected BEL ratio has the usual chi-square limiting calibration under appropriate weak-dependence conditions. The PBEL literature uses a degrees-of-freedom adjustment that accounts for the blocking construction. That chi-square calibration is a diagnostic, not the production robustness claim.

1.2 Bartlett correction — corrected mathematical contract

Kitamura's result is not the simple i.i.d. empirical-likelihood factor. The general correction factor contains third- and fourth-order cumulant/moment tensors of the normalized dependent block estimating functions, plus derivative terms for a general smooth-function parameter. The corrected statistic multiplies (or equivalently rescales) the BEL likelihood-ratio by that factor to the order used in Kitamura's expansion.

The important higher-order result: the uncorrected BEL error is $O(1/n)$; the Bartlett-corrected version improves this to $O(1/n^2)$ under the paper's conditions. Kitamura states that the correction can be estimated from sample block moments or by bootstrap.

Important DCM correction

Do not take the familiar i.i.d. scalar EL formula and silently use it for overlapping BEL. That formula is valid for the ordinary scalar empirical-likelihood mean; the general multivariate dependent-block version is different.

```
BARTLETT_IID_EL
    permitted for independent-cluster EL diagnostics

BARTLETT_BEL
    separate estimator
    dependent-block cumulants required
    RESEARCH_ONLY until coverage validation

Preference: test both a direct block-moment plug-in estimator and a
bootstrap-estimated Bartlett factor in the coverage tournament.
High-order moment estimation is itself noisy in the small samples
where the correction is most attractive.
```

1.3 Bartlett does not replace fixed-b

These address different approximation errors. Small-b BEL assumes the block fraction $b = L/n \to 0$ (with $L \to \infty$). Kitamura's higher-order analysis considers a block-growth regime of that type. Fixed-b instead holds $b \to b$■ $\in (0,1]$. The limiting distribution is then nonstandard but pivotal and depends on the fixed block fraction / kernel. Critical values become those of the fixed-b limit, not χ^2. Simulation studies in the fixed-b BEL literature found that this calibration better approximated the finite-sample BEL distribution than standard chi-square.

Method	Correct role
BEL_CHI2	Diagnostic
BEL_BARTLETT	Higher-order small-b research only
BEL_FIXED_B	Main temporal robustness challenger
ABEL	Convex-hull robustness
PBEL	Block-choice robustness challenger

Table 1. Phase A method roles. These roles stay frozen.

1.4 Convex-hull caveat

A Bartlett correction improves the approximation of the distribution conditional on BEL being properly defined. It does not eliminate the convex-hull problem. BEL coverage can remain severely bounded below nominal levels when dependence is strongly positive, block size is large relative to sample size, or moment dimension grows. ABEL, fixed-b, and coverage diagnostics retain separate jobs. That keeps Phase A frozen rather than reopening it.

2. Block-length selection policy

No universal optimum exists. The block length optimal for long-run variance estimation is not guaranteed to be optimal for BEL confidence coverage. Literature commonly quotes orders such as L ~ n^{1/3} or, for other distributional problems, nearby polynomial rates. Recent BEL discussions note that optimal coverage-based block selection is not fully solved and that practitioners often borrow selectors from the block-bootstrap or spectral-density literature.

2.1 Politis–White as a candidate generator, not a theorem

For the score-difference time series, estimate the autocovariance sequence with a flat-top kernel, form the long-run variance, and apply the corrected Politis–White constant (2009 correction fixed the stationary-bootstrap variance constant while preserving the optimal-block formulas). For circular / moving blocks the usual plug-in form is the sibling estimator.

DCM must not say “Politis–White L* therefore optimal BEL block length.” Instead: Politis–White supplies a center for a candidate grid. The actual BlockPlan is chosen by the finite-sample coverage harness on training-only data. A sensible research candidate family is a small set of multiples of L* subject to a minimum valid block count and regime boundaries. Those constants are a research grid, not a production theorem. For fixed-b, evaluate a frozen training-only grid of b ratios and store the critical-value table by b. PBEL remains useful because it avoids choosing one constant block length; its blocks grow through an arithmetic progression and simulations have shown increased stability to the block-selection problem.

2.2 BlockPlan three new fields

```
block_selection_objective
block_selector_version
block_selector_training_hash

Example payload:
  block_selection_objective = COVERAGE_AND_TYPE1_ERROR
  block_length_rule         = POLITIS_WHITE_CENTERED_GRID
  selector_training_cutoff  = 202X-XX-XX
  coverage_target           = 0.95
  type1_error_target        = 0.05

Selection minimizes a preregistered training-only criterion
(coverage error + Type-I deviation). That objective is DCM
engineering, not a claimed universal block-selection theorem.
```

2.3 Hard invariants on blocks

- M ≥ 1, L ≥ 1, and Q = ■(N − M)/L■ + 1 (or the overlapping analogue) must produce enough valid blocks for the method.
- training_cutoff < first_evaluation_timestamp.
- No block may cross regime_id.
- time_unit ∈ {SLATE, DAY, EVENT_WINDOW}.

3. Phase B — EventWorld hierarchy and PrimitiveStatLedger

This is the major engineering target. v5.4.1 already contained the correct invariant: simulate an event once and derive related markets from the same world while respecting conservation and cross-market identities. v6 formalizes that as a primitive event-state ledger that feeds an exact, versioned PrizePicks settlement engine.

3.1 Two-level world objects

Do not store the entire simulation definition redundantly inside every Monte Carlo world. EventWorldSet is one frozen simulation run for one real sporting event (seeds, evidence hash, parameter snapshot, world_count, primitive_schema_version). EventWorld is the indexed draw: latents, team states, player states, primitive_ledger_hash, valid flag, and invariant_failures. A world should ordinarily be generated so structural violations are impossible. valid=False is mainly defensive auditing.

3.2 Event layer

For event e, frozen pre-cutoff evidence is hashed. Sample a shared event state that can contain pace/tempo, game competitiveness, scoring environment, weather, officiating regime, venue, team-strength state and other sport-relevant latents. Discrete uncertain conditions are represented explicitly (lineup/availability regimes when genuine pregame uncertainty remains). Do not force every sport into one universal vector with meaningless null fields internally; EventLatentState is an envelope around sport-specific latent payloads.

3.3 Discrete uncertain regimes

DiscreteEventRegime carries regime_type, regime_value, prior_probability, and evidence_ids. Examples: STARTER_ACTIVE / STARTER_LIMITED / STARTER_OUT; ROOF_OPEN / ROOF_CLOSED; NORMAL_ROTATION / SHORT_ROTATION. Uncertainty about these states is simulated rather than converted into one arbitrary deterministic penalty.

3.4 Team layer and player opportunity vs efficiency

The team layer generates the total opportunity pool before players receive shares: possessions, offensive plays, pass attempts, rush attempts, plate appearances, shots, shifts / time-on-ice, rallies / points, or fight-time states. Player opportunity and player efficiency stay separate objects (OpportunityState, EfficiencyState). Player opportunity shares obey conservation for any exhaustively allocated team resource. A logistic-normal or Dirichlet-style allocation is appropriate in many aggregate models; count allocations can use multinomial / Dirichlet-multinomial mechanisms when overdispersion matters.

This split makes later variance attribution possible: Var(Y) decomposes into opportunity, efficiency, event, parameter, and interaction components — not necessarily as a simple additive analytic identity, but through world-level attribution stored per market. That is how DCM explains why a line is fragile.

3.5 Primitive stats, not props, are the source of truth

This is the most important Phase-B design choice. A world must not independently draw Points, Rebounds, Assists, PRA, and Fantasy Score. Draw the primitive sporting outcomes and calculate everything else. For world w, every market is f(ledger_w; definition_version). Identities such as PRA = PTS + REB + AST must hold exactly in every world, not merely on average. Derived fantasy scores, H+R+RBI, passing+rushing combinations, goalie saves, combos, and other composites come from the same primitive state. This prevents impossible simulations such as a player's component totals contradicting their composite projection.

3.6 Event path versus aggregate simulator

Phase B is hybrid. The full event state does not need possession-by-possession simulation for every market. Any market whose dependence structure requires event ordering should come from a path model: next action given current event state. Basketball can use possession or stint states; football drives/plays; baseball plate appearances / base-out state; tennis point/game/set; MMA round / fight-duration. Aggregate models remain for markets where an event path adds little incremental information. That keeps runtime tractable.

3.7 Hierarchical shrinkage lives inside the simulator

Pregame player parameters are not raw recent averages. A generic hierarchical parameter shrinks small samples toward team / role / league priors. Simulation uncertainty then naturally contains parameter uncertainty plus process noise. That decomposition is stored per market.

3.8 Cross-event card worlds

Same-event correlation is solved by the shared event simulation. For different events, couple independently by default. A deterministic low-discrepancy / permutation coupling may be used only when there is evidence for a genuine global slate factor. Do not assume event-simulation row 417 from Game A has a meaningful relationship with row 417 from Game B.

4. Stat semantics, MarketDefinitionRegistry, conservation tests

4.1 Semantic types

```
class StatSemanticType(Enum):
    PRIMITIVE      = "PRIMITIVE"
    DERIVED        = "DERIVED"
    COMPOSITE      = "COMPOSITE"
    PLATFORM_SCORE = "PLATFORM_SCORE"

If semantic_type == PRIMITIVE: formula must be null.
If semantic_type != PRIMITIVE: value comes only from the definition engine.
No simulator method may write a derived value into the primitive ledger.
```

4.2 MarketDefinitionRegistry

Extends v5.4.1 rather than replacing it. Registry key is exactly (Platform, League, Market, DefinitionVersion). No fuzzy fallback. Fields include output_unit, source_stat_keys, formula, overtime_policy, push_policy, participation_policy_version, reboot_policy_version, authoritative_source_ids, verified, verification_hash.

4.3 WorldProjectionResult

Converts primitive worlds into prop-space worlds: computed_value plus component_values plus computation_hash. Audits can prove that a PRA distribution came from the same points / rebounds / assists worlds. Every world must satisfy the identity exactly.

4.4 Conservation rules are first-class objects

Not ad-hoc assertions. ConservationRule has rule_id, sport, league, rule_type ∈ {SUM_EQUALS, COMPONENT_IDENTITY, RESOURCE_ALLOCATION, BOUND, LOGICAL}, expression, tolerance, rule_version. InvariantResult records passed, observed, expected, residual. Phase B acceptance requires P(structural invariant failure) = 0 for deterministic identities. The simulator should reject a world, or preferably make it impossible to generate one, when a structural identity fails.

Basketball examples (not the only identities)

- Regulation team player-minutes ≈ 240 per team in the NBA; each overtime adds 25 team player-minutes.
- Fully allocated opportunity pools sum to the team pool.
- Team points equal the sum of player points under the chosen scoring definition.
- 2PA = FGA – 3PA; FGM = 2PM + 3PM; REB = OREB + DREB; PTS = 2·2PM + 3·3PM + FTM.
- Made ≤ attempts on every attempt family.

5. Basketball primitive registry — first live implementation

Implemented against the accepted development baseline (canonical v5.4.1 untouched; source and ledger hashes verified against the install manifest). The basketball primitive layer is executable, not merely architectural.

Family	Keys
Opportunity / volume	Minutes, FGA, 3PA, 2PA, FTA

Family	Keys
Makes	3PM, 2PM, FGM, FTM
Box primitives	OREB, DREB, REB, AST, STL, BLK, TO, PTS
Derived / combo (not independently sampled)	PRA, PR, PA, RA, Blocks+Steals, Fantasy

Table 2. Basketball primitive topology.

- Fail closed: supplying contradictory primitives raises instead of allowing an internally inconsistent world.
- Shared-world derivation: event_sim.py samples the necessary stochastic components; the registry derives dependent statistics and composites.
- Every currently production-supported NBA/WNBA basketball market in the v5.4.1 capability registry has a primitive binding.
- Distribution semantics: basketball forecasting obtains its market family from the registry instead of the old hand-maintained COUNT_MARKETS set. That closes the gap where FGA / FTA / 3PA were supported but unrecognized by that list.
- Separation preserved: primitive validity ≠ PrizePicks selectability. Platform formulas, push, OT, DNP/Reboot, and exact settlement remain under the verified Stat Definition layer.
- Connected to Phase B/C via phase_bc_conservation_rules(), phase_bc_primitive_values(), build_phase_bc_primitive_ledger().
- Executable conservation rules cover 2PA, FGM, REB, PTS, PRA, PR, PA, RA, Blocks+Steals, made≤attempt, and team-minute conservation.

Verification snapshot from the development patch: Phase B/C + basketball/GPLVM targeted 23/23 PASS; v5.3 + v5.4 + v5.4.1 relevant regressions 96 OK, 1 skip; entire runnable suite 316 OK, 1 skip. Six TestV510ExactHAR methods still require two external historical HAR files that are not in the runtime. That FileNotFoundError is pre-existing and is not introduced by the schema or basketball patches.

6. Planned sport registries — football, baseball, UFC

These four sports were chosen because they expose four different event structures and will stress-test whether the schemas are genuinely universal. Football is the next implementation target after basketball.

Sport	Path unit	Opportunity pools	Representative primitives	Hard identities
Basketball (LIVE)	Possession / stint	Minutes, FGA, FTA, 3PA	2PM/3PM/FTM, OREB/DREB, AST, STL, BLK, TO	Shot-clock identities; 240 + 25·OT team-minutes; PTS formula
Football (NEXT)	Drive / play	Snaps, dropbacks, rush attempts, targets, routes	Pass yds/att/TD/INT, rush yds/att/TD, rec yds/rec/TD, sacks	Team plays = pass + rush + sacks taken; rec ≤ targets; TD accounting
Baseball	PA / base-out	PA, AB, BF, outs, innings	H, 2B, 3B, HR, BB, K, R, RBI, IP, ER, R	AB + BB + HBP + SF + SH + CI = PA; H = 1B+2B+3B+HR; TB identity
UFC / MMA	Round / fight-time	Fight seconds, significant-strike attempts	Strikes landed/attempted, TD, ctrl time, KD	Landed ≤ attempted; time ≤ scheduled; decision vs finish exclusivity

Table 3. Multi-sport primitive plan. New sports must not invent new ledger/world/settlement schemas.

Football will be the hardest conservation test: play-level accounting, sack classification, target-vs-reception, rushing+receiving dual-role players, defensive exclusion from many PrizePicks reboot paths, and kicker/punter zero-opportunity special cases already encoded in the 2026 Reboot table.

7. Phase C — EntryContract and decomposed PrizePicks settlement

PrizePicks' 2026 Player Picks system is no longer adequately described by a simple fixed Flex/Power payout table. Current cards can produce both a Leaderboard Win and a Minimum Guarantee; the member receives the higher amount, not the sum. Leaderboard scoring depends on projection type. Minimum Guarantee payouts depend on the card result. Actual structures may differ by lineup, promotions, same-game combinations and special projections. PrizePicks tells users to rely on the payout displayed for the submitted lineup. Therefore Phase C freezes an EntryContract; it does not infer payout from card size.

7.1 EntryContract and EntryPickContract

EntryContract freezes platform, platform_rule_version, submitted_at, entry_type, stake, currency, picks, minimum_guarantee_definition_id, leaderboard_definition_id, payout_display_hash, contract_hash. The payout shown at submission is immutable. Do not later reconstruct it from today's rules.

Each EntryPickContract freezes projection_id, player, team, event, market_definition_id, line, side ∈ {MORE, LESS}, modifier ∈ {STANDARD, DEMON, GOBLIN, OTHER}, offered_side_verified, leaderboard_point_weight, reboot_rule_version, participation_rule_version. Inherited invariant: GOBLIN → analytics/settlement allowed, selection forbidden.

7.2 Three settlement dimensions (never collapse to one enum)

Dimension	States	Job
AdministrativeState	ACTIVE, DNP, REBOOT, CANCELLED, INVALID_MARKET, UNRESOLVED	Was the pick removed or rewritten by platform administration?
ComparisonState	WIN, LOSS, TIE, NOT_APPLICABLE	Did the official stat beat, miss, or push the frozen line?
EconomicState	COUNTS_AS_WIN, COUNTS_AS_LOSS, TIER_REDUCTION, REMOVED, UNRESOLVED	How does this pick enter payout-tier and eligibility math?

Table 4. Decomposed pick state. Physical comparison ≠ pick result ≠ lineup economics.

A final active pick fails closed if its declared Higher/Lower comparison or WIN/LOSS/PUSH result disagrees with its authoritative actual value, frozen line, and direction. If authoritative settlement cannot be established the state is UNRESOLVED, not a guessed WIN/LOSS. That preserves the canonical fail-closed settlement doctrine.

7.3 Two different card counts

This mathematical distinction is essential. The payout tier generally steps downward for both voids and ties, but the eligibility population is not the same. Tied picks are not removed for same-team / minimum-pick eligibility rules; DNP / Reboot picks are. PrizePicks explicitly distinguishes these effects. Settlement code cannot execute effective_picks = original_picks − voids and use one number for everything. LineupSettlement therefore stores original_pick_count, administrative_removed_count, tie_count, payout_tier_count, eligibility_population_count, and distinct_remaining_team_count as separate fields. They are never recomputed later from one generic effective_pick_count.

There are also explicit 2-pick tie special cases in the official rules (one win plus one tie pays a listed amount; one loss plus one tie loses). Those are versioned table rows, not inferred arithmetic.

7.4 Exact pick-state function

An offered selection is the frozen tuple of player, market definition, line, side, modifier, and platform-rule version. Ordinary stat comparison is only the inner kernel. Actual settlement is administrative state first, then comparison if still ACTIVE, because DNP, Reboot, cancellation, board error and participation thresholds can override an ordinary comparison.

8. Reboot, DNP, same-team refund, official-score timing

8.1 Versioned Reboot rule tables

Reboot mechanics are sport- and board-specific. One global Boolean is forbidden. The first frozen snapshot is PRIZEPICKS_PLAYER_PICKS_2026-08-25_V1. Implementation lives as reboot.py plus rules/prizepicks/{NBA,WNBA,NFL,CFB,MLB}_dated snapshots.

Sport	v6 Reboot implementation (snapshot V1)
NBA	MORE + full-game + eligible stat; player leaves in 1H and does not return in 2H. If projection already achieved before leaving, normal settlement applies.
WNBA	MORE + full-game regular-season + eligible stat; leaves 1H; no 2H return.
NFL	MORE + full-game regular/postseason; no preseason; offensive eligibility; K/P special zero-opportunity path; defense excluded. Combo Squares can reboot when a member qualifies.
CFB	MORE + full-game, but explicit frozen player_reboot_eligible registry required. No broad event-phase assumption hard-coded (Playoff language vs live qualifying-player list).
MLB	MORE + full-game batter; leaves with ≤2 plate appearances; pitchers and MLBLIVE excluded. Plate appearances, not at-bats.

Table 5. Reboot snapshot. LESS selections are not Rebooted.

Two fail-closed decisions are intentional. For NBA/WNBA/NFL, PrizePicks says some “X achievements in X attempts” and partial-period markets are excluded but does not publish a complete machine-readable stat-key list on the public page. v6 therefore requires stat_reboot_eligible = VERIFIED_TRUE/FALSE from a frozen registry. Unknown means UNRESOLVED, not guessed eligible.

8.2 Same-team refund

After DNP/Reboot removal, if all remaining active players are from the same team, current rules say the lineup becomes ineligible and is refunded. Formally: after administrative removals define the remaining active set; if that set is nonempty, same-team, and the condition arose through DNP/Reboot reversion, the platform contract returns a refund rather than a win. This is an economic state on the lineup, not a comparison rewrite on the surviving picks.

8.3 Sport truth vs platform truth

PrizePicks states that settlement is based on official league / trusted-provider results as originally posted at conclusion; later statistical corrections generally do not alter an already settled pick, although PrizePicks can correct clear platform scoring errors. DCM therefore stores OfficialSportStat (authoritative_timestamp, provider_id, revision_number) separately from PickSettlement (official_result_timestamp, platform_settlement_timestamp). If those later differ, SPORT_STAT_REVISION / SPORT_STAT_CORRECTION must not automatically be classified as a forecasting error.

9. Leaderboard, Minimum Guarantee, and unmodelled group EV

9.1 Leaderboard scoring is not 1/0

As of the August 24, 2026 Potential Outcomes page: a winning Demon scores 1.05; a winning Standard scores 1.00; a winning Goblin, Discounted, or Stack scores 0.95. A winning selection that belongs to a same-player stack scores 0.95 regardless of what it otherwise would have scored. Losses, ties, DNPs, Reboots and unsettled picks after an Early Exit score zero.

Lineup score $S(C) = \Sigma\, q_j \cdot 1\{R_j = \text{WIN}\}$. If multiple members tie for the highest group score, the Leaderboard payout L (the amount frozen from the submitted lineup) is divided among those winners: $R_{LB} = L / T$ if $S_u = \max S$, else 0.

9.2 Never reconstruct L from a generic table

PrizePicks publishes standard Leaderboard rates (validation only; e.g. published illustrations have included 6-pick Power Leaderboard at 25× and 6-pick Flex at 12.5×) but explicitly says some lineups receive different rates because of discounted projections or multiple selections from the same team/game. displayed_leaderboard_payout is captured at submission and stored as an immutable EntryContract field. "6 picks → assume 12.5×" is forbidden.

9.3 Final payout is a max, not a sum

If a lineup qualifies for both Leaderboard and Minimum Guarantee, the member receives the higher amount; the payouts do not stack. R_final = max(R_LB, R_MG). Early Payout changes Minimum Guarantee eligibility while a High Score / Leaderboard result can still remain relevant. Those interactions are encoded in choose_final_player_picks_payout(...).

9.4 Consequence: Leaderboard EV is initially unmodelled

DCM can calculate the Minimum Guarantee exactly from simulated card worlds and the frozen contract. It cannot truthfully calculate E[R_LB] unless it also knows or models the distribution of other contestants' group scores. Therefore production reporting is:

```
MINIMUM_GUARANTEE_RETURN:   MODELED
LEADERBOARD_RETURN:         UNMODELED_UNLESS_GROUP_DISTRIBUTION_AVAILABLE
TOTAL_PLATFORM_EV:          LOWER_BOUND_OR_PARTIAL
```

This is a major improvement over the earlier "6-leg parlay" objective. P(payout > 0) and P(net > 0) remain explicit metrics; they are not substitutes for the economic return distribution. A standard 6-flex 4/6 result can pay a positive amount that is still a losing ticket relative to stake. "Received a payout" is not the same as "did not lose money."

9.5 Research-only tie-aware IID Leaderboard helper

For a future group-score model, suppose the user's simulated score is s, there are n–1 opponents, p_s = P(S = s), q_s = P(S < s). If exactly t opponents tie the user and all others finish below, the payout is L/(t+1). The expectation is L · Σ binom(n–1, t) p_s^t q_s^{n–1–t} / (t+1). Closed form: L · ((q_s + p_s)^n – q_s^n) / (n p_s) when p_s > 0, and L · q_s^{n–1} when p_s = 0. Modelling only P(S_u = max S) and multiplying by full L overstates EV whenever ties occur. Implemented as expected_leaderboard_return_iid(...) and marked RESEARCH_ONLY because actual assigned group entries cannot yet be assumed IID.

10. Frozen schema objects, hash lineage, failure codes

10.1 Object inventory (PHASE_BC_SCHEMA_V1_2026-08-25)

The module provides deeply immutable contracts for: BlockPlan, EventWorldSet / EventWorld, EventLatentState, DiscreteEventRegime, PlayerWorldState, OpportunityState, EfficiencyState, PrimitiveStatLedger / PrimitiveStatEntry, StatDefinition, MarketDefinition, ConservationRule / InvariantResult, WorldProjection / WorldProjectionResult, EntryPickContract, EntryContract, PickSettlement, LineupSettlement, WorldLineupOutcome. Nested mappings are deep-frozen so a stat map cannot be changed after its parent object was hashed. A machine-readable freeze lives at schemas/Phase_BC_Immutable_Contracts.json. A unit test compares that JSON with runtime introspection so changing a dataclass field without updating the immutable contract breaks the suite.

Schema contract SHA-256

```
6e78dacc19843338643bdcabc7477fd3ce2dd065da1e9629646dacc21cdb1f22
```

10.2 Hash lineage (machine-enforced, no skipped stages)

```
EvidenceGraphHash
     ↓
```

```
ParameterSnapshotHash
      ↓
EventWorldSetHash
      ↓
PrimitiveLedgerHash
      ↓
MarketDefinitionHash
      ↓
WorldProjectionHash
      ↓
EntryContractHash
      ↓
SettlementRuleHash
      ↓
WorldLineupOutcomeHash

A lineage stage cannot be skipped.
An already-populated hash cannot be silently replaced.
If a historical result changes on replay, DCM identifies the layer.
```

10.3 Standardized Phase B/C failure codes

None of these silently become a probability adjustment. They are state / integrity failures.

Code	Meaning
EVENT_WORLD_INVALID	World failed generation or validity audit
PRIMITIVE_CONSERVATION_FAILURE	Ledger violates a deterministic identity
DERIVED_IDENTITY_FAILURE	Composite disagrees with components
UNVERIFIED_MARKET_DEFINITION	Market not in verified registry
UNIT_MISMATCH	Unit disagreement across definition / ledger / line
UNKNOWN_PLATFORM_RULE	No exact payout-table / rule match
UNKNOWN_REBOOT_RULE	Sport/board/stat reboot eligibility unknown
UNKNOWN_PARTICIPATION_RULE	Participation threshold unknown
ENTRY_CONTRACT_INCOMPLETE	Missing frozen payout display or pick fields
PLATFORM_SETTLEMENT_UNRESOLVED	Authoritative settlement not established
LEADERBOARD_EV_UNIDENTIFIED	Group-score distribution unavailable
SPORT_STAT_REVISION_AFTER_SETTLEMENT	League correction after platform settle

Table 6. Fail-closed codes.

10.4 LineupSettlement and WorldLineupOutcome accounting

A final lineup must account for every frozen pick: active + void/DNP + reboot = frozen contract pick count, and wins + losses + pushes = active picks. WorldLineupOutcome stores per-world win_count, loss_count, tie_count, hypothetical_minimum_return, leaderboard_score, hypothetical_total_return, net_profit. From M worlds DCM estimates $P(K = k)$, $P(K \geq 5)$ for a six-leg card, $P(\Pi > 0)$, and $P(\Pi - \text{stake} > 0)$ as distinct metrics. leaderboard_return_status ∈ {MODELED, OBSERVED, UNMODELED, NOT_APPLICABLE}. settlement_status ∈ {FINAL, REFUND, PARTIAL, UNRESOLVED}.

11. Research challengers — DPMM and Bayesian GP-LVM

11.1 Dirichlet process mixture models

A standard DPMM is $G \sim DP(\alpha, G_0)$, $\theta_i \mid G \sim G$, $Y_i \mid \theta_i \sim K(\cdot \mid \theta_i)$, with predictive density the kernel mixed over G. Ferguson established the DP; Sethuraman supplied stick-breaking: $V_k \sim Beta(1, \alpha)$, $\pi_k = V_k \prod_{\blacksquare<k}(1 - V_\blacksquare)$, $\theta_k^* \sim G_0$, $G = \Sigma\, \pi_k\, \delta_{\theta_k^*}$. Under the Chinese-restaurant representation the expected number of occupied components grows roughly logarithmically, so DCM is not forced to pre-declare “three regimes” or “five regimes.” Neal’s MCMC (including auxiliary-variable methods for nonconjugate kernels) is the operational reference; Escobar–West is the density-estimation reference.

DPMM does not replace Game → Team → Opportunity → Efficiency. The interesting challenger is hierarchical structural model + DPMM residual distribution. After known role/context effects, ε_i is mixed rather than assumed $N(0, \sigma^2)$. That can capture asymmetric tails, unanticipated multimodality, heterogeneous variances, latent efficiency regimes, and unusual role-comparable subpopulations. Opportunity itself stays conservation-constrained. Construction: $r_i^{eff} \sim$ DPMM, then $\eta_i = \eta_i^{structural} + r_i^{eff}$. Multivariate related efficiency dimensions can mix $N(\mu, \Sigma)$ over G and discover residual regimes with different covariance structures.

```
DPMM_EFFICIENCY_RESIDUAL_V1
STATUS = SHADOW_CHALLENGER
PRODUCTION_AUTHORITY = FALSE

PROMOTION REQUIRES:
    chronological CRPS improvement
  + LogS / Brier noninferiority where applicable
  + joint ES / VS safety where multivariate
  + subgroup safety
  + stability across future slates

Operational rule: use only the posterior predictive.
Inferred component labels are not semantic truths.
"Cluster 3" must never become a hard DCM rule.
Production replay freezes a posterior sample bank + hash;
it does not launch unconstrained MCMC on every board run.
```

11.2 Bayesian GP-LVM (and deferred GPDM)

GP-LVM models high-dimensional observations through a nonlinear low-dimensional latent representation — a probabilistic nonlinear extension of PCA. The Bayesian variational version (Titsias & Lawrence) integrates uncertainty over the latent inputs, addresses overfitting, and allows ARD to help determine useful latent dimensionality. Sparse GP-LVM is the practical direction once basketball history is large. GPflow remains a realistic implementation path.

GP-LVM is not used to directly generate Higher/Lower probabilities. Best first use: a nonlinear role/context-state representation over minutes, usage, shot mix, assists, rebounding involvement, pace, availability, starter state, opponent context and rest. Latent coordinates then challenge the current OOD / role-state machinery.

First encoded experiment: variational sparse Bayesian GP-LVM + ARD, candidate latent dimensions 2–5, chronologically trained preprocessing, comparisons against PCA, probabilistic PCA and no latent embedding. Promotion requires future-only improvement in proper scoring, calibration and ranking — not reconstruction quality or attractive clusters. Hard production veto until then.

Second-stage variant only if latent role states show meaningful sequential structure: Gaussian Process Dynamical Models add GP dynamics in latent space rather than treating games as exchangeable. That is relevant to bench → starter → injury-limited → normalized starter. Test ordinary Bayesian GPLVM first; GPDM only if temporal dynamics add incremental value.

12. Engineering sequence, promotion gates, and non-goals

12.1 Revised sequence (accepted)

Step	Work	Gate
1	Lock missing BEL formulas; BlockPlan candidate generator + coverage-driven selector; Bartlett RESEARCH_ONLY	Phase A roles unchanged; coverage harness exists
2	EventWorld → PrimitiveStatLedger → MarketDeriver with mandatory conservation tests (basketball LIVE)	P(structural failure)=0 on deterministic identities
3	Football primitive registry against the same schema (then baseball, UFC)	No new ledger/world/settlement types invented
4	PrizePicks EntryContract + PlatformSettlementAdapter on current official rules	Exact reboot/DNP/tie/MG/LB; fail closed on unknown
5	Card optimizer around the true return function; retain P(K=k) and P(Π>0) as explicit metrics	Does not treat those as the economic contract

Table 7. Build order.

12.2 Production vs research authority

Component	Authority
v5.4.1 event-first / unit-safe / versioned definitions	PRODUCTION (inherited)
PHASE_BC_SCHEMA_V1_2026-08-25	FROZEN CONTRACT
Basketball PrimitiveRegistry + conservation	DEVELOPMENT BASELINE, executable
PrizePicks rule snapshot 2026-08-25_V1	DEVELOPMENT BASELINE (13 settlement tests reported)
BEL_CHI2 / ABEL / PBEL / BEL_FIXED_B roles	PRODUCTION ROLES FROZEN; implementations as previously gated
BEL_BARTLETT	RESEARCH_ONLY
expected_leaderboard_return_iid	RESEARCH_ONLY
DPMM_EFFICIENCY_RESIDUAL_V1	SHADOW_CHALLENGER
Bayesian GP-LVM + ARD experiment	RESEARCH_ONLY, hard production veto
GPDM	DEFERRED
Leaderboard EV without group distribution	UNMODELED — report as such

Table 8. Authority matrix.

12.3 Explicit non-goals

- Do not reopen Phase A production gates to insert Bartlett.
- Do not treat Politis–White L* as optimal BEL block length.
- Do not sample composites independently of primitives.
- Do not infer PrizePicks payout from card size.
- Do not collapse DNP / Reboot / Tie into one effective_pick_count.
- Do not classify post-settlement sport-stat revisions as forecast errors by default.
- Do not promote DPMM cluster labels into business rules.
- Do not use GP-LVM reconstruction quality as a production argument.
- Do not claim TOTAL_PLATFORM_EV is fully modelled while Leaderboard group scores are unknown.
- Do not modify the canonical v5.4.1 installation package.

13. Object-model chain and world-level metrics

```
PREDICTION CHAIN
```

```
FrozenEvidence
      ↓
EventWorldSet
      ↓
EventWorld
      ↓
PrimitiveStatLedger
      ↓
MarketDefinitionRegistry
      ↓
WorldProjectionResult

SETTLEMENT CHAIN (independent translation)
EntryContract + PlatformRuleVersion + WorldProjectionResult
      ↓
RebootRuleRegistry + Participation rules
      ↓
WorldPickState / PickSettlement
      ↓
WorldLineupOutcome / LineupSettlement
      ↓
MinimumGuaranteeReturn
      ↓
LeaderboardReturn (if modelable)
      ↓
EconomicReturnDistribution = max(LB, MG) under frozen display

AFTER GAMES
OfficialSportStat + PlatformSettlementEvidence
      ↓
PickSettlement → LineupSettlement
```

13.1 Metrics that are allowed to be reported as modelled

- $P\blacksquare(K = k) = (1/M)\ \Sigma\ 1\{K_w = k\}$ and tail probabilities such as $P\blacksquare(K \geq 5)$ on a six-leg card.
- Exact Minimum Guarantee return distribution under the frozen contract.
- $P\blacksquare(\Pi > 0)$ and $P\blacksquare(\Pi - \text{stake} > 0)$ — different from each other and from $P(K \geq \text{threshold})$.
- Component / identity audits: $PRA_w = PTS_w + REB_w + AST_w$ exactly.
- Conservation residual distribution (should be a point mass at 0 for deterministic rules).
- Variance attribution across opportunity / efficiency / event / parameter when stored.

13.2 Metrics that must be labelled partial or unmodelled

- Leaderboard return without a group-score distribution.
- Total platform EV when only MG is identified.
- Any EV that reconstructs L from a published standard table instead of the frozen display.
- Any settlement that required an unknown reboot or participation fact.

14. Appendix — schema field inventory and implementation notes

14.1 BlockPlan fields

```
schema_version, plan_id, learning_revision
time_unit, sport_family, league, market_family
training_start, training_cutoff, regime_id, regime_break_rule_version
method ∈ {BEL_CHI2, BEL_FIXED_B, ABEL, PBEL, BEL_BARTLETT}
block_length_rule, block_lengths_evaluated, selected_block_length
overlap, block_start_spacing, fixed_b_ratio
block_selection_objective, block_selector_version, block_selector_training_hash
critical_value_version, nominal_alpha, plan_hash
```

14.2 EventWorldSet / EventWorld / ledger

```
EventWorldSet:
  world_set_id, run_id, event_id, sport, league
  event_start, forecast_cutoff
  learning_revision, model_version, league_rule_version
  evidence_graph_hash, parameter_snapshot_hash
  simulation_method, world_count
  seed_definition_version, seed_material_hash
  primitive_schema_version, worlds_hash

EventWorld:
  world_set_id, world_index
  event_latents, team_states, player_states
  primitive_ledger_hash, valid, invariant_failures

PrimitiveStatEntry:
  entity_type ∈ {EVENT, TEAM, PLAYER, PAIR}
  entity_id, team_id, stat_key, value, unit
  primitive_definition_version
```

14.3 Settlement objects

```
PickSettlement:
  official_stat_value, comparison_line
  administrative_state, comparison_state, economic_state
  participation_evidence_ids, official_stat_source_ids
  official_result_timestamp, platform_settlement_timestamp

LineupSettlement:
  original_pick_count, administrative_removed_count, tie_count
  payout_tier_count, eligibility_population_count
  distinct_remaining_team_count
  minimum_guarantee_multiplier / return
  leaderboard_score / return / return_status
  final_platform_return, net_return, settlement_status
```

14.4 Implementation notes from this design cycle

- Canonical v5.4.1 package remains untouched. All work is on a reconstructed development copy whose source/ledger hashes were verified against the install manifest before modification.
- Basketball is the first actual Event world → immutable primitive ledger → conservation checks → market-definition boundary implementation against the universal contract.
- PrizePicks rule layer and Leaderboard split math were implemented in a development module with a reported 13-test suite plus a research-only IID Leaderboard helper and test.
- Deep freeze goes beyond ordinary frozen=True dataclasses: nested mappings cannot be mutated after hash.
- No payout-table nearest-match. Unknown rule → UNKNOWN_PLATFORM_RULE.
- Next high-value build: football primitive registry, then wire WorldProjection + EntryContract through the existing Reboot / MG / Leaderboard adapter so one basketball (then football) world produces a full WorldLineupOutcome.

14.5 Closing statement

The most important discovery of this cycle is that Phase C must not optimize a generic 6-leg payout table. The current PrizePicks contract has Minimum Guarantee, Leaderboard scoring, variable submitted multipliers, reversion mechanics, and sport-specific administrative settlement. The v6 simulator generates the sporting world once. The settlement engine translates each world through the exact frozen platform contract. That is the cleanest possible separation between sports prediction and platform economics.

End of DCM v6.0.0 Phase A/B/C Master Blueprint. Document ID DCM-V6-BLUEPRINT-2026-08-27. Schema freeze PHASE_BC_SCHEMA_V1_2026-08-25. Canonical v5.4.1 untouched.